

	FATE MONTHLY REPORT Doc.No : FATE-001 Version : 1.0 Date : 02-05-2024	
---	--	---

FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE

Performance assessment
Annex
April 2024



Contents

0.1	NIGHT1: Contingency tables	2
0.2	DAY1: Contingency tables	13
0.3	NIGHT2: Contingency tables	18
0.4	DAY2: Contingency tables	22
0.5	NIGHT3: Contingency tables	24
0.6	DAY3: Contingency tables	28

wind speed ms^{-1}	Observations		
	$ws < 4.2$	$4.2 < ws < 7.6$	$ws > 7.6$
Model data	$ws < 4.2$	3650	1372
	$4.2 < ws < 7.6$	2038	2538
	$ws > 7.6$	214	1049
Number of points 16241 ; PC=64.3%; EBD=1.9%; POD1=61.8%; POD2=51.2%; POD3=79.2%			

Table 1: night1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}	Observations		
	$ws < 4.2$	$4.2 < ws < 7.6$	$ws > 7.6$
Model data	$ws < 4.2$	5488	406
	$4.2 < ws < 7.6$	414	4288
	$ws > 7.6$	0	265
Number of points 16241 ; PC=91.8%; EBD=0.0%; POD1=93.0%; POD2=86.5%; POD3=95.5%			

Table 2: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

0.1 NIGHT1: Contingency tables

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

wind speed ms^{-1}	Observations		
	ws < 12.0	12.0 < ws < 18.0	ws > 18.0
Model data	ws < 12.0	14473	141
	12.0 < ws < 18.0	84	1272
	ws > 18.0	0	26
Number of points 16241 ; PC=98.1%; EBD=0.0%; POD1=99.4%; POD2=88.4%; POD3=76.7%			

Table 3: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)
 ms^{-1}

FATE - Monthly Report

wind direction degree		North	East	South	Observations	West
Model data	North	5901	399	45		88
	East	526	802	423		5
	South	62	278	3138		266
	West	212	60	83		198

Number of points 12486 ; PC=80.4%; EBD=1.4%; POD1=88.1%; POD2=52.1%; POD3=85.1%; POD4=35.5%

Table 4: night1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	6599	128	0		16
	East	92	1337	50		0
	South	0	74	3633		28
	West	10	0	6		513

Number of points 12486 ; PC=96.8%; EBD=0.0%; POD1=98.5%; POD2=86.9%; POD3=98.5%; POD4=92.1%

Table 5: night1 - Contingency table for wind direction (degree) processed with AR (1H)

FATE - Monthly Report

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	2008	611
	7.1 < rh < 15.3	2109	3768
	rh > 15.3	474	1655
Number of points 16143 ; PC=64.3%; EBD=3.1%; POD1=43.7%; POD2=62.4%; POD3=83.5%			

Table 6: night1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	4171	229
	7.1 < rh < 15.3	419	5622
	rh > 15.3	1	183
Number of points 16143 ; PC=93.7%; EBD=0.0%; POD1=90.9%; POD2=93.2%; POD3=96.6%			

Table 7: night1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent	Observations		
	rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	15596	37
	50.0 < rh < 70.0	28	323
	rh > 70.0	0	7
Number of points 16143 ; PC=99.5%; EBD=0.0%; POD1=99.8%; POD2=88.0%; POD3=97.4%			

Table 8: night1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	2612	538	2
	1.9 < pwv < 3.5	550	3132	858
	pwv > 3.5	8	669	6901
Number of points 15270 ; PC=82.8%; EBD=0.1%; POD1=82.4%; POD2=72.2%; POD3=88.9%				

Table 9: night1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	3096	109	0
	1.9 < pwv < 3.5	74	4163	72
	pwv > 3.5	0	67	7689
Number of points 15270 ; PC=97.9%; EBD=0.0%; POD1=97.7%; POD2=95.9%; POD3=99.1%				

Table 10: night1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	2372	1415	446
	0.6 < see < 0.8	821	1125	1089
	see > 0.8	157	504	1152
Number of points 9081 ; PC=51.2%; EBD=6.6%; POD1=70.8%; POD2=37.0%; POD3=42.9%				

Table 11: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	3312	504	24
	0.6 < see < 0.8	380	2215	267
	see > 0.8	11	352	2444
Number of points 9081 ; PC=83.8%; EBD=0.4%; POD1=89.4%; POD2=72.1%; POD3=89.4%				

Table 12: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	3223	0	316
	0.6 < see < 0.8	0	2953	316
	see > 0.8	127	91	2371
Number of points 9081 ; PC=94.1%; EBD=4.9%; POD1=96.2%; POD2=97.0%; POD3=79.0%				

Table 13: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	3695	4	21
	0.6 < see < 0.8	0	3051	20
	see > 0.8	8	16	2714
Number of points 9081 ; PC=99.5%; EBD=0.3%; POD1=99.8%; POD2=99.3%; POD3=98.5%				

Table 14: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.24

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1094	378	8
	$3.4 < \tau < 5.8$	489	1151	501
	$\tau > 5.8$	82	1063	3347
Number of points 8113 ; PC=68.9%; EBD=1.1%; POD1=65.7%; POD2=44.4%; POD3=86.8%				

Table 15: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1551	139	0
	$3.4 < \tau < 5.8$	114	2216	202
	$\tau > 5.8$	0	237	3654
Number of points 8113 ; PC=91.5%; EBD=0.0%; POD1=93.2%; POD2=85.5%; POD3=94.8%				

Table 16: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 0.0

wavefront coherence time ms		Observations		
		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1452	20	172
	$3.4 < \tau < 5.8$	131	1995	172
	$\tau > 5.8$	82	577	3684
Number of points 8113 ; PC=87.9%; EBD=3.1%; POD1=87.2%; POD2=77.0%; POD3=91.5%				

Table 17: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

wavefront coherence time ms		Observations		
		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1663	1	9
	$3.4 < \tau < 5.8$	2	2556	9
	$\tau > 5.8$	0	35	3847
Number of points 8113 ; PC=99.4%; EBD=0.1%; POD1=99.9%; POD2=98.6%; POD3=99.5%				

Table 18: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1427	1369
	0.6 < glf < 0.7	752	1069
	glf > 0.7	273	353
Number of points 8228 ; PC=38.0%; EBD=17.2%; POD1=58.2%; POD2=38.3%; POD3=21.1%			

Table 19: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2138	413
	0.6 < glf < 0.7	306	1972
	glf > 0.7	8	406
Number of points 8228 ; PC=80.5%; EBD=0.4%; POD1=87.2%; POD2=70.7%; POD3=84.3%			

Table 20: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2149	377
	0.6 < glf < 0.7	30	2395
	glf > 0.7	273	19
Number of points 8228 ; PC=76.5%; EBD=18.3%; POD1=87.6%; POD2=85.8%; POD3=44.2%			

Table 21: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2438	17
	0.6 < glf < 0.7	6	2760
	glf > 0.7	8	14
Number of points 8228 ; PC=99.1%; EBD=0.5%; POD1=99.4%; POD2=98.9%; POD3=97.9%			

Table 22: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

wind speed ms^{-1}	Observations		
	$ws < 4.2$	$4.2 < ws < 7.6$	$ws > 7.6$
Model data	$ws < 4.2$	1749	381
	$4.2 < ws < 7.6$	2169	4359
	$ws > 7.6$	121	1870
Number of points 15770 ; PC=65.1%; EBD=1.1%; POD1=43.3%; POD2=65.9%; POD3=81.2%			

Table 23: day1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}	Observations		
	$ws < 4.2$	$4.2 < ws < 7.6$	$ws > 7.6$
Model data	$ws < 4.2$	3556	331
	$4.2 < ws < 7.6$	483	5930
	$ws > 7.6$	0	349
Number of points 15770 ; PC=90.5%; EBD=0.0%; POD1=88.0%; POD2=89.7%; POD3=93.5%			

Table 24: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

0.2 DAY1: Contingency tables

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

wind speed ms^{-1}	Observations		
	ws < 12.0	12.0 < ws < 18.0	ws > 18.0
Model data	ws < 12.0	13804	146
	12.0 < ws < 18.0	114	1217
	ws > 18.0	0	33
Number of points 15770 ; PC=97.7%; EBD=0.0%; POD1=99.2%; POD2=87.2%; POD3=85.1%			

Table 25: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)
 ms^{-1}

FATE - Monthly Report

wind direction degree		North	East	South	Observations	West
Model data	North	4277	352	16		97
	East	80	241	73		0
	South	12	38	2133		340
	West	1966	16	269		3799

Number of points 13709 ; PC=76.2%; EBD=0.3%; POD1=67.5%; POD2=37.2%; POD3=85.6%; POD4=89.7%

Table 26: day1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	6004	57	0		200
	East	49	573	8		0
	South	0	17	2426		38
	West	282	0	57		3998

Number of points 13709 ; PC=94.8%; EBD=0.0%; POD1=94.8%; POD2=88.6%; POD3=97.4%; POD4=94.4%

Table 27: day1 - Contingency table for wind direction (degree) processed with AR (1H)

FATE - Monthly Report

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	659	372
	7.1 < rh < 15.3	1339	4572
	rh > 15.3	222	1944
Number of points 15748 ; PC=70.9%; EBD=1.4%; POD1=29.7%; POD2=66.4%; POD3=89.4%			

Table 28: day1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	1966	170
	7.1 < rh < 15.3	254	6493
	rh > 15.3	0	225
Number of points 15748 ; PC=94.7%; EBD=0.0%; POD1=88.6%; POD2=94.3%; POD3=97.2%			

Table 29: day1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent	Observations		
	rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	15183	20
	50.0 < rh < 70.0	28	284
	rh > 70.0	0	9
Number of points 15748 ; PC=99.5%; EBD=0.0%; POD1=99.8%; POD2=90.7%; POD3=92.0%			

Table 30: day1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	1597	370	3
	1.9 < pwv < 3.5	760	3262	691
	pwv > 3.5	35	694	7786
Number of points 15198 ; PC=83.2%; EBD=0.3%; POD1=66.8%; POD2=75.4%; POD3=91.8%				

Table 31: day1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	2293	84	0
	1.9 < pwv < 3.5	99	4182	100
	pwv > 3.5	0	60	8380
Number of points 15198 ; PC=97.7%; EBD=0.0%; POD1=95.9%; POD2=96.7%; POD3=98.8%				

Table 32: day1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

wind speed ms^{-1}	Observations		
	$ws < 4.2$	$4.2 < ws < 7.6$	$ws > 7.6$
Model data	$ws < 4.2$	3380	1300
	$4.2 < ws < 7.6$	2076	2502
	$ws > 7.6$	246	1004
Number of points 15598 ; PC=63.5%; EBD=2.2%; POD1=59.3%; POD2=52.1%; POD3=79.0%			

Table 33: night2 - Contingency table for wind speed (ms^{-1}) in standard configuration

0.3 NIGHT2: Contingency tables

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	5665	418	66
	East	542	727	617
	South	41	255	2845
	West	103	79	82

Number of points 11952 ; PC=78.3%; EBD=1.6%; POD1=89.2%; POD2=49.2%; POD3=78.8%; POD4=24.4%

Table 34: night2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	1680	534
	7.1 < rh < 15.3	2103	3255
	rh > 15.3	615	1879

Number of points 15476 ; PC=60.2%; EBD=4.3%; POD1=38.2%; POD2=57.4%; POD3=81.0%

Table 35: night2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	2518	587
	1.9 < pwv < 3.5	560	2611
	pwv > 3.5	19	872

Number of points 14823 ; PC=80.6%; EBD=0.4%; POD1=81.3%; POD2=64.2%; POD3=89.1%

Table 36: night2 - Contingency table for precipitable water vapor (mm) in standard configuration

FATE - Monthly Report

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	2537	1442	578
	0.6 < see < 0.8	995	1310	1145
	see > 0.8	246	703	1377
Number of points 10333 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 37: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	3592	0	395
	0.6 < see < 0.8	0	3295	395
	see > 0.8	186	160	2705
Number of points 10333 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 38: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time		Observations		
ms		tau < 3.4	3.4 < tau < 5.8	tau > 5.8
Model data	tau < 3.4	1110	373	25
	3.4 < tau < 5.8	562	1264	487
	tau > 5.8	97	1325	4054
Number of points 9297 ; PC=69.1%; EBD=1.3%; POD1=62.7%; POD2=42.7%; POD3=88.8%				

Table 39: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		tau < 3.4	3.4 < tau < 5.8	tau > 5.8
Model data	tau < 3.4	1514	39	207
	3.4 < tau < 5.8	158	2004	207
	tau > 5.8	97	919	4359
Number of points 9297 ; PC=84.7%; EBD=3.3%; POD1=85.6%; POD2=67.7%; POD3=91.3%				

Table 40: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1443	1566
	0.6 < glf < 0.7	981	1286
	glf > 0.7	242	359
Number of points 9454 ; PC=36.8%; EBD=16.6%; POD1=54.1%; POD2=40.0%; POD3=20.9%			

Table 41: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2390	384
	0.6 < glf < 0.7	34	2803
	glf > 0.7	242	24
Number of points 9454 ; PC=77.3%; EBD=18.0%; POD1=89.6%; POD2=87.3%; POD3=44.3%			

Table 42: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	1771	403	69
	4.2 < ws < 7.6	2178	4286	1098
	ws > 7.6	94	1796	3944
Number of points 15639 ; PC=63.9%; EBD=1.0%; POD1=43.8%; POD2=66.1%; POD3=77.2%				

Table 43: day2 - Contingency table for wind speed (*ms⁻¹*) in standard configuration

0.4 DAY2: Contingency tables

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	4007	349	19
	East	130	176	70
	South	16	30	1955
	West	2067	39	471

Number of points 13554 ; PC=73.1%; EBD=0.5%; POD1=64.4%; POD2=29.6%; POD3=77.7%; POD4=89.3%

Table 44: day2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	509	276
	7.1 < rh < 15.3	1445	3916
	rh > 15.3	303	2608

Number of points 15614 ; PC=65.4%; EBD=1.9%; POD1=22.6%; POD2=57.6%; POD3=88.2%

Table 45: day2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	1680	474
	1.9 < pwv < 3.5	830	2829
	pwv > 3.5	37	926

Number of points 15096 ; PC=79.3%; EBD=0.3%; POD1=66.0%; POD2=66.9%; POD3=89.6%

Table 46: day2 - Contingency table for precipitable water vapor (mm) in standard configuration

wind speed ms^{-1}	Observations		
	$ws < 4.2$	$4.2 < ws < 7.6$	$ws > 7.6$
Model data	$ws < 4.2$	3246	1383
	$4.2 < ws < 7.6$	2120	2403
	$ws > 7.6$	313	985
Number of points 15598 ; PC=61.1%; EBD=3.0%; POD1=57.2%; POD2=50.4%; POD3=75.3%			

Table 47: night3 - Contingency table for wind speed (ms^{-1}) in standard configuration

0.5 NIGHT3: Contingency tables

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	5631	475	81
	East	541	710	632
	South	57	244	2831
	West	108	50	104

Number of points 11960 ; PC=77.8%; EBD=1.7%; POD1=88.9%; POD2=48.0%; POD3=77.6%; POD4=27.4%

Table 48: night3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	1507	627
	7.1 < rh < 15.3	2210	3241
	rh > 15.3	681	1867

Number of points 15476 ; PC=57.2%; EBD=4.7%; POD1=34.3%; POD2=56.5%; POD3=76.7%

Table 49: night3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	2603	824
	1.9 < pwv < 3.5	506	2225
	pwv > 3.5	57	1019

Number of points 14823 ; PC=76.3%; EBD=1.0%; POD1=82.2%; POD2=54.7%; POD3=85.5%

Table 50: night3 - Contingency table for precipitable water vapor (mm) in standard configuration

FATE - Monthly Report

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	2444	1515	560
	0.6 < see < 0.8	1075	1247	1214
	see > 0.8	209	690	1345
Number of points 10299 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 51: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	3553	0	399
	0.6 < see < 0.8	0	3343	399
	see > 0.8	175	109	2720
Number of points 10299 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 52: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time		Observations		
ms		tau < 3.4	3.4 < tau < 5.8	tau > 5.8
Model data	tau < 3.4	1160	345	16
	3.4 < tau < 5.8	551	1197	554
	tau > 5.8	103	1378	4000
Number of points 9304 ; PC=68.3%; EBD=1.3%; POD1=63.9%; POD2=41.0%; POD3=87.5%				

Table 53: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		tau < 3.4	3.4 < tau < 5.8	tau > 5.8
Model data	tau < 3.4	1532	35	218
	3.4 < tau < 5.8	179	2010	218
	tau > 5.8	103	875	4352
Number of points 9304 ; PC=84.8%; EBD=3.5%; POD1=84.5%; POD2=68.8%; POD3=90.9%				

Table 54: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1335	1455
	0.6 < glf < 0.7	1057	1437
	glf > 0.7	249	340
Number of points 9461 ; PC=36.6%; EBD=16.8%; POD1=50.5%; POD2=44.5%; POD3=19.4%			

Table 55: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2339	321
	0.6 < glf < 0.7	53	2886
	glf > 0.7	249	25
Number of points 9461 ; PC=77.8%; EBD=18.0%; POD1=88.6%; POD2=89.3%; POD3=44.4%			

Table 56: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	1780	461	76
	4.2 < ws < 7.6	2074	4279	1091
	ws > 7.6	161	1753	3964
Number of points 15639 ; PC=64.1%; EBD=1.5%; POD1=44.3%; POD2=65.9%; POD3=77.3%				

Table 57: day3 - Contingency table for wind speed (ms^{-1}) in standard configuration

0.6 DAY3: Contingency tables

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	3938	366	37
	East	109	163	92
	South	13	30	1918
	West	2221	41	468

Number of points 13576 ; PC=70.8%; EBD=0.7%; POD1=62.7%; POD2=27.2%; POD3=76.3%; POD4=85.9%

Table 58: day3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	357	369
	7.1 < rh < 15.3	1578	3839
	rh > 15.3	305	2609

Number of points 15614 ; PC=63.1%; EBD=2.0%; POD1=15.9%; POD2=56.3%; POD3=86.2%

Table 59: day3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	1634	502
	1.9 < pwv < 3.5	919	2791
	pwv > 3.5	34	876

Number of points 15143 ; PC=76.9%; EBD=0.7%; POD1=63.2%; POD2=66.9%; POD3=86.0%

Table 60: day3 - Contingency table for precipitable water vapor (mm) in standard configuration